

Should Platforms Be Allowed to Sell on Their Own Marketplaces?

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PART 1

Research Question

Presenter: Jinghao Mu

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Should Platforms Be Allowed to Sell on Their Own Marketplaces?

Hagiu, Teh, and Wright (2022)

Motivation: A Dual Role platform

Many large platforms now play two roles at the same time.
(Eg. Amazon, JD.com, Apple App Store...)

- They run a marketplace for third-party sellers.
- They also sell their own products on the same marketplace.

The key tension

The platform is both the market **organizer** and one of the **sellers** in the market.



Why Dual Mode Is Controversial

Dual mode has both potential harms and potential efficiency benefits.

Regulatory concerns

- **Product imitation:** copy successful sellers using marketplace data.
- **Self-preferring:** steer consumers toward the platform's own products.
- **Innovation harm:** weaken third-party sellers' investment incentives.

Potential benefits

- **One-stop shopping:** consumers can compare more options in one place.
- **Efficient supply:** the platform may be a better seller in some categories.
- **Price pressure:** the platform's product can constrain third-party prices.

Policy question

Should regulators ban dual mode itself, or target specific harmful conduct?

Research Question

The paper asks three linked questions.

1. Does **banning** dual mode always **raise consumer surplus and welfare**?
2. If dual mode is banned, does the platform **switch to seller mode or marketplace mode**?
3. Are targeted **rules on imitation and self-preferring** better than a full ban?

Key Insight

The problem is not dual mode itself, but how the platform uses its power.

- A ban on dual mode is not always good for consumers or welfare.
- Rules that target imitation and self-preferring can work better.

Location and Contribution

The paper is located at the intersection of platform economics and antitrust policy.

- It studies a platform that can choose among **marketplace mode**, **seller mode**, and **dual mode**.
- It connects the policy debate on platform bans with a formal welfare analysis.
- It separates **dual mode itself** from two possible bad conducts: **product imitation** and **self-preferring**.



Presentation Roadmap

The presentation is organized into five parts.

1. Part 1: Motivation and main question
2. Part 2: Baseline model setup
3. Part 3: Baseline equilibria and price squeeze
4. Part 4: Ban on dual mode
5. Part 5: Imitation, self-preferring, and policy comparison

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PART 2

Baseline Model

Presenter: Jialong Zheng

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Players I: Consumers and Competitive Benchmark

Consumers

- Outside option of not buying anything (e.g., postpone purchase or buy another product): $v_o \sim G^a$
- Platform transaction benefit: $b > 0$ (finding products, one-click payment, delivery tracking, easy returns).
- Choice rule: buy the option with the highest net utility.

^a G is log-concave and continuously differentiable on $[0, \bar{v}_o]$, with density g .

Fringe sellers

- At least two identical sellers.
- Sell a basic product with value v .
- Consumers already know them and can buy directly.
- Bertrand competition gives the price benchmark.^a

^aAll marginal costs are normalized to zero.

Players II: Platform and Innovative Seller

Innovative (superior) seller S

- Benefits from innovation; discovered only if listed on M .
- Better product value: $\nu + \Delta > \nu > 0$.
- Innovation: chooses quality investment Δ .

Platform M

- Chooses marketplace, seller, or dual mode.
- Seller: sells its own offering, valued at $\nu + \sigma$.
- Marketplace: sets commission $\tau \geq 0$ for third-party sellers.



Values and Parameters

Product values are:

Product	Consumer value	Meaning
Fringe sellers	v	Basic product
Innovative seller S	$v + \Delta$	Better product
Platform M	$v + \sigma$	Platform's own product

- $b > 0$: convenience benefit from buying through the platform.
- $v_o \sim G$: outside option.
- τ : platform commission.

Intuition

$\sigma > 0$ means the platform is a better seller than fringe sellers; $\sigma < 0$ means the opposite.



Innovation by Seller S

Seller S chooses innovation $\Delta \geq \Delta^l$ and pays fixed cost $K(\Delta)$.

$$K'(\bar{\Delta}) = G(\nu + \sigma + b)$$

Meaning of $\bar{\Delta}$

$\bar{\Delta}$ is the high innovation level that can arise when the market gives strong rewards to innovation.

Intuition

The left side is marginal cost of innovation. The right side is the demand gain from better quality.



Product Discovery

Consumers do not know about S at the start.

- Consumers know the fringe sellers.
- Consumers may know the platform's own product.
- Consumers discover S only if S is available on the platform.

Why this matters

The platform is not only a sales channel. It is also a discovery channel.

Intuition

If a policy changes the platform's mode, it may also change whether consumers find S .

Three Platform Modes

Mode	What M does	Key feature
Marketplace	Hosts third-party sellers	Earns commission τ
Seller	Sells its own product	No marketplace for S
Dual	Hosts sellers and sells own product	Both roles at once

Intuition

The policy debate is about whether the third mode should be allowed.



Timing

The game is solved by backward induction.

1. M chooses its mode and sets commission τ if there is a marketplace.
2. S decides whether to join the platform and chooses Δ .
3. Sellers set prices.
4. Consumers choose what to buy.

Equilibrium concept

Subgame Perfect Nash Equilibrium.



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PART 3

Equilibrium Analysis

Presenter: Binghui Guo

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Marketplace Mode

In marketplace mode, M only charges commission τ .

$$p_i \leq \tau + \Delta, \quad p_o \leq \tau + \Delta - b$$

Showrooming constraint

If $\tau > b$, seller S can use the platform for discovery and then sell directly.

Intuition

The platform can charge at most the extra convenience value b that it gives consumers.

Marketplace Equilibrium

Proposition 1:

$$\tau^{mkt} = b, \quad G(v) = K'(\Delta^{mkt}), \quad p_i^* = b + \Delta^{mkt}$$

$$\Pi^{mkt} = bG(v), \quad \pi^{mkt} = \Delta^{mkt}G(v) - K(\Delta^{mkt})$$

Intuition

M sets the highest fee that does not push trade outside the platform. S invests until marginal cost equals marginal market gain.

Seller Mode

In seller mode, the platform only sells its own product.

$$\max_{p_m \leq b + \sigma} p_m G(v + b + \sigma - p_m) \quad (3.1)$$

The competitive constraint binds:

$$p_m^* = b + \sigma$$

Proposition 2

M sells to all consumers, while S sells to no one.

Intuition

The platform captures its convenience value and its product advantage, but consumers do not discover S .

Dual Mode: Two Pricing Cases

In dual mode, M both charges commission and sells its own product.

Case 1: $\sigma \geq \Delta$

The platform's own product is good enough. Consumers buy from M .

$$p_i^* = \tau, \quad p_m^* = \tau + \sigma - \Delta$$

Case 2: $\Delta > \sigma$

Seller S has the better product. But M still puts pressure on S 's price.

Intuition

Dual mode matters most when S is better, but M can still discipline S 's price.

Price Squeeze

In the price squeeze equilibrium:

$$p_i^* = p_m^* + \Delta - \sigma$$

$$p_m^* \in [\max\{\tau - \Delta + \sigma, 0\}, \tau + \min\{\sigma, 0\}]$$

$$\Pi = \tau G(\nu + \sigma + b - p_m^*)$$

Intuition

M may not sell, but its own product caps S 's price. A lower S price raises demand and commission income.

Innovation Constraint

Define $\bar{\tau}$ by:

$$(\bar{\Delta} - \sigma - \bar{\tau})G(\nu + b + \sigma) - K(\bar{\Delta}) = 0$$

Lemma 1:

- If $\tau \leq \bar{\tau}$, then S chooses high innovation: $\Delta = \bar{\Delta}$.
- If $\tau > \bar{\tau}$, then S chooses low innovation: $\Delta = \Delta^l$.

Intuition

A high commission lowers S 's margin. If the margin is too low, innovation is not worth the cost.

Dual Mode Equilibrium

Proposition 3: M balances two limits.

- Showrooming constraint: $\tau \leq b$.
- Innovation constraint: $\tau \leq \bar{\tau}$.

If high innovation is profitable enough:

$$\tau^{dual} = \min\{b, \bar{\tau}\}, \quad \Delta^{dual} = \bar{\Delta}$$

Otherwise:

$$\tau^{dual} = b, \quad \Delta^{dual} = \Delta^l$$

Intuition

The platform may charge less to keep S willing to innovate, because innovation expands the market.



Which Mode Does the Platform Choose?

Corollary 1:

- M prefers marketplace mode over seller mode iff $\sigma \leq 0$.
- Dual mode is always better for M than marketplace mode.
- There is a threshold $\underline{\sigma} > 0$ such that:

$$\Pi^{dual} > \Pi^{sell} \quad \text{iff} \quad \sigma < \underline{\sigma}$$

Intuition

Dual mode creates more transactions through price squeeze. But if M 's own product is much better, M may prefer to sell alone.

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PART 4

Imitation and Self-Preferencing

Presenter: Zhiyuan Chen

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What Is Added?

This part adds two practices to the baseline dual-mode model: **product imitation** and **Self-preferencing**.

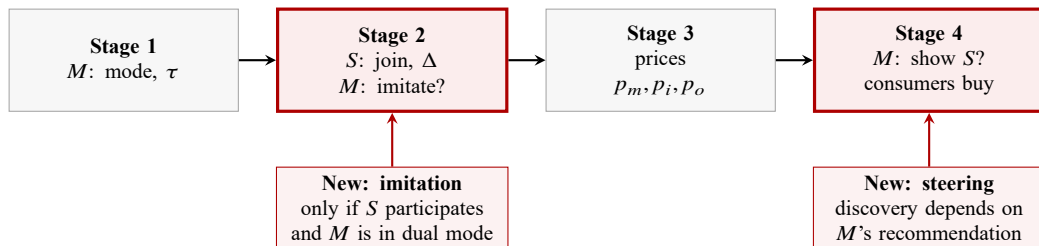
Product imitation

- **Assumption:** copied product gives surplus $v + \Delta$.
- **Choice:** S : participate or not; then M : imitate or not if S participates.
- **Motivation:** marketplace data helps M identify and copy successful innovations.

Self-preferencing

- **Assumption:** consumers discover S through M 's recommendation.
- **Choice:** M : recommend S or not after prices are set.
- **Motivation:** consumers often rely on platform search, ranking, or recommendations to find new products.

Timing: Where Section 4 Changes the Game



Changes from the baseline

- M copying choice: M imitates or not.
- M disclosure choice: M shows S or not.

Marketplace and Seller Mode

Marketplace mode: same commission τ from any marketplace sale.

If M shows S

Buy S through the marketplace iff:

$$\Delta + b - p_i \geq \max\{\Delta - p_o, b - \tau, 0\} \quad (4.1)$$

- Left: surplus from S on M .
- Right: other best outside option.

If M hides S

Buy a fringe seller on the marketplace iff:

$$b - \tau \geq 0 \quad (4.2)$$

- Left: surplus from a fringe seller on M .
- Right: no purchase.

Recommendation rule

- Show S if Eq. (4.1), holds: M earns commission τ .
- Hide S if Eq. (4.1), fails but Eq. (4.2), holds.

Seller mode is unchanged: M sells only its own product.

Dual Mode: Recommendation Decision

What does M compare?

- Own expected margin from selling M 's product or imitation.
- Expected commission from showing S .

Recommendation rule

- M choose to hide S
 - M 's expected margin $>$ expected commission from S ;
 - Or the marketplace-purchase condition for S , Eq. (4.1), fails.
- M choose to show S
 - M 's expected margin $<$ expected commission from S ;
 - and the marketplace-purchase condition for S , Eq. (4.1), holds.

Pricing Without Imitation

Without imitation, the subgame can be exploitative or price-squeezing.

1. **Exploitative:** M hides S , and charges the highest price allowed by outside options.
2. **Price squeeze:** M shows S , but caps S 's inside price and margin.

Exploitative equilibrium

- **Exploitative price:**

$$p_m^* = \min\{\tau, b\} + \sigma \quad (4.3)$$

- M vs fringe through marketplace: $\nu + \sigma + b - p_m^* > \nu + b - \tau \Rightarrow p_m^* < \tau + \sigma$
- M vs fringe not through marketplace: $\nu + \sigma + b - p_m^* > \nu \Rightarrow p_m^* < b + \sigma$
- **Exploitative profit:** using the seller-mode profit function, Eq. (3.1), and Eq. (4.3):

$$\Pi_{no-imi}^{exploit} = (\min\{\tau, b\} + \sigma)G(\nu + b - \min\{\tau, b\}). \quad (4.4)$$

Pricing Without Imitation: Price Squeeze

Without imitation, products and quality gaps are the same as in the baseline. The only new instrument is steering:

- If S is shown, pricing follows the baseline price-squeeze logic.
- Self-preferencing adds one more constraint: M must prefer showing S to hiding it.

Pricing With Imitation

Imitation matters when S 's product is better than M 's original product:

$$\Delta > \sigma. \quad (4.5)$$

Exploitative equilibrium with imitation

M sells the copied product to all consumers at

$$p_m^* = \min\{\tau, b\} + \Delta. \quad (4.6)$$

Price-squeeze equilibrium with imitation

If it exists, homogeneous-product competition implies

$$p_i^* = p_m^* = \tau. \quad (4.7)$$

Mechanism

Imitation extracts innovation surplus; steering prevents S from using discovery to discipline the platform.



Proposition 5: Dual Mode Equilibrium

With product imitation and self-preferencing, S always chooses minimum innovation:

$$\Delta^{dual} = \Delta^l. \quad (4.8)$$

Case	Equilibrium conduct	M 's profit
$\sigma \geq \Delta^l$	no imitation; M sells	$(b + \sigma)G(\nu)$
$\sigma < \Delta^l$	high fee or imitation	$(b + \Delta^l)G(\nu)$

Summary

$$\Pi^{dual} = (b + \max\{\sigma, \Delta^l\})G(\nu). \quad (4.9)$$



Why Innovation Collapses

In every relevant dual-mode configuration, S 's equilibrium operating profit is zero.

- If M imitates, the copied product captures the innovation surplus.
- If M does not imitate because $\sigma \geq \Delta$, M already has the better product.
- If M does not imitate because τ is high, the commission extracts S 's margin.

$$\pi_S(\Delta) = -K(\Delta) \quad \Rightarrow \quad \Delta = \Delta^l. \quad (4.10)$$

Economic intuition

Imitation and steering remove both rewards from innovation: product advantage and access to consumers.

Proposition 6: Ban Dual Mode

Because dual mode is weakly best for M , a ban forces M into a pure mode.

Region	New mode	Π	π	CS	Δ, W
$\sigma \geq \Delta^I$	Seller	=	=	=	$\Delta =, W =$
$0 < \sigma < \Delta^I$	Seller	↓	=	=	$\Delta =, W \downarrow$
$\sigma \leq 0$	Marketplace	↓	↑	=	$\Delta \uparrow, W \uparrow$

Difference from the baseline

With perfect steering, consumer surplus is already low in dual mode, so banning dual mode no longer changes CS.

Why the Dual-Mode Ban Can Still Lower Welfare

The empirically relevant case is often $0 < \sigma < \Delta^I$.

- M has a selling advantage relative to fringe sellers: $\sigma > 0$.
- S 's innovative product is still superior: $\Delta^I > \sigma$.
- After the ban, M switches to seller mode.
- Consumers cannot combine S 's better product with the platform benefit b .

Policy lesson

A structural ban does not restore innovation or effective competition; it only removes the marketplace channel from M 's choice set.

Behavioral Remedies

Section 4 then compares three conduct remedies with the structural ban.

Remedy	Direct target	Restored mechanism
Ban self-preferencing	display choice	on-platform competition
Ban imitation	copying choice	innovation incentives
Ban both	display and copying	competition and innovation

Main question

Can targeted conduct rules preserve the useful parts of dual mode while removing the harmful practices?

Ban Self-Preferencing Only

M must always show S when S is listed.

- The exploitative no-display equilibrium disappears.
- The showrooming constraint on τ is restored.
- Product imitation is still allowed.

Post-intervention dual mode

$$\tau = b, \quad \Delta = \Delta^l, \quad \Pi = (b + \max\{\sigma - \Delta^l, 0\})G(\nu + \Delta^l), \quad \pi = 0. \quad (4.11)$$

Intuition

Display neutrality restores price competition, but copying still destroys S 's innovation incentive.

Proposition 7: Effects of Banning Self-Preferencing

There is a cutoff

$$\underline{\sigma}^{steer} \in (0, \min\{\Delta^l, \underline{\sigma}\}] \quad (4.12)$$

such that:

Region	M 's mode	Π	π	CS	Δ, W
$\sigma \geq \Delta^l$	Seller	=	=	=	$\Delta =, W =$
$\underline{\sigma}^{steer} < \sigma < \Delta^l$	Seller	↓	=	=	$\Delta =, W \downarrow$
$\sigma \leq \underline{\sigma}^{steer}$	Dual	↓	=	↑	$\Delta =, W \uparrow$

Interpretation

If M remains in dual mode, consumers gain from lower prices and higher demand.

Ban Imitation Only

Banning imitation is equivalent to letting M credibly commit not to copy.

Innovation-restoring condition

If

$$b + \max\{\sigma, \Delta^I\} \leq \bar{\tau} \quad (4.13)$$

or

$$\bar{\tau} G(\nu + \sigma + b) \geq (b + \max\{\sigma, \Delta^I\}) G(\nu), \quad (4.14)$$

then M sets $\tau = \bar{\tau}$ and S chooses $\Delta = \bar{\Delta}$.

Intuition

A no-copying commitment can benefit M when larger innovation expands demand enough to offset a less extractive commission.



Proposition 8: Effects of Banning Imitation

M always continues in dual mode after this remedy.

Condition	Π	π	CS	Δ	W
$b + \max\{\sigma, \Delta^I\} > \bar{\tau}$ and (14) fails	=	=	=	=	=
$b + \max\{\sigma, \Delta^I\} \leq \bar{\tau}$ or (14) holds	↑	=	↑	↑	↑

Mechanism

When the ban matters, equilibrium moves from Δ^I to $\bar{\Delta}$ and demand rises from $G(\nu)$ to $G(\nu + \sigma + b)$.



Ban Both Practices

If imitation and self-preferencing are both banned, the dual-mode equilibrium becomes the baseline dual-mode equilibrium from Proposition 3.

- Showing S restores on-platform price competition.
- No-copying restores S 's innovation incentives.
- The showrooming constraint again limits τ .

Synergy

The condition for innovation to increase is less restrictive than under a ban on imitation alone, because self-preferencing no longer weakens the showrooming constraint.

Proposition 9: Effects of Banning Both

The post-remedy mode depends on the same cutoff $\underline{\sigma}$ from the baseline.

Region	M 's mode	Π	π	CS	Δ, W
$\sigma \geq \max\{\underline{\sigma}, \Delta^I\}$	Seller	=	=	=	$\Delta =, W =$
$\underline{\sigma} < \sigma < \Delta^I$	Seller	↓	=	=	$\Delta =, W \downarrow$
$\sigma \leq \underline{\sigma}$	Dual	↓	↑ if $b < \bar{\tau}$	↑	$\Delta \uparrow$ if $b < \bar{\tau}$ or (8); $W \uparrow$

Takeaway

When M stays in dual mode, banning both practices raises consumer surplus and welfare by restoring the baseline dual-mode mechanisms.

A Data-Sharing Alternative

The paper also discusses a possible rule requiring M to share third-party seller data with all third parties.

- This removes M 's exclusive imitation advantage.
- But now all fringe sellers can imitate S .
- S 's innovation incentive is again eliminated.

Comparison

In the model, data sharing is weakly worse than banning both imitation and self-preferencing because it does not restore S 's incentive to innovate.



Corollary 2: Behavioral Remedies vs. Dual-Mode Ban

First consider $\sigma > 0$, so a dual-mode ban makes M choose seller mode.

Remedy	Relevant region	Π	π	CS	Δ, W
Ban self-preferencing	$\sigma \leq \underline{\sigma}^{steer}$	$\uparrow$	$=$	$\uparrow$	$\Delta =, W \uparrow$
Ban self-preferencing	$\sigma > \underline{\sigma}^{steer}$	$=$	$=$	$=$	$\Delta =, W =$
Ban imitation	all $\sigma > 0$	$\uparrow$	$=$	$\uparrow$	$\Delta \uparrow, W \uparrow$
Ban both	$\sigma \leq \underline{\sigma}$	$\uparrow$	$\uparrow$	$\uparrow$	$\Delta \uparrow, W \uparrow$

Message for empirically relevant cases

Conduct remedies weakly dominate the structural ban, and often strictly improve outcomes.

Corollary 2 When $\sigma \leq 0$

If $\sigma \leq 0$, a dual-mode ban makes M choose marketplace mode.

Remedy	Always weakly higher	Innovation	Welfare
Ban self-preferencing	π , CS	lower than marketplace	up iff condition (9) holds
Ban imitation	π , CS	up iff $b + \Delta' \leq \bar{\tau}$ or (14)	up iff Δ rises
Ban both	π , CS	up iff $b \leq \bar{\tau}$ or (8)	up iff Δ rises or (9) holds

Why ambiguity returns

When fringe sellers have the selling advantage, marketplace mode can be a strong benchmark; conduct remedies need enough innovation or demand gains to dominate it.

Section 4 Takeaways

1. Perfect imitation and perfect steering make dual mode weakly best for M .
2. In dual mode, S chooses Δ^I because innovation rents are extracted.
3. A dual-mode ban does not directly fix copying or steering.
4. Banning self-preferencing restores price competition but not innovation.
5. Banning imitation restores innovation when commitment is valuable enough.
6. Banning both practices restores the baseline dual-mode mechanisms.

Main policy message

The harmful conduct is often a better regulatory target than the platform's dual role itself.

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